

Project II - Pandas & Matplotlib: Occupancy Estimation

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1. Project Overview

The aim of this project is to practice data manipulation and visualization using Python's Pandas and Matplotlib libraries. The project uses a real dataset, exploring its structure, cleaning missing values, performing grouping and filtering operations, and producing a variety of visualizations.

2. Dataset

"Occupancy_Estimation_1_new.csv" dataset contains sensor readings from a room along with the corresponding occupancy count at each timestamp. Sensors measure temperature, light, sound, CO2, and motion. The goal of the dataset is to estimate the number of people in the room from these sensor readings.

Properties:

Number of rows	10,129	Total observations
Number of columns	19	Total variables
Total missing values	57,671	NaN count across all columns
Date range	2017/12/22 onward	Recording period
Target column	Room_Occupancy_Count	Values: 0, 1, 2 or 3 persons

Data Type	Date	Time	S1_Temp, S2_Temp, S3_Temp, S4_Temp	S1_Light, S2_Light, S3_Light, S4_Light	S1_Sound, S2_Sound, S3_Sound, S4_Sound	S5_CO2	S5_CO2_Slope	S6_PIR, S7_PIR	Room_Occupancy_Count
Column Name	object	object	float	float/int	float	float	float	float	float

3. Questions

1. Load the dataset using Pandas
2. Display the first 8, 12 and 25 rows separately
3. Check the data type of at least 5 rows separately
4. Check the data type of the whole columns
5. List the columns names of the data frame
6. Copy your data frame to a new data frame and:
 - a. Change the data type of columns whose data type is integer into float
 - b. Change the data type of columns whose data type is float into integer

7. Create a new data frame the consists of columns which are not numeric.
8. Display those whose value is >, >=, <, <=, == a specific column's value (for example, salary>3000)
9. Perform groupings (at least 4 kinds), based on criteria. The grouping shall have aggregation (for example, average, maximum, minimum etc.)
10. Display range of rows (1-5) and first, second, fourth, sixth columns.
11. Display third through tenth rows and the first four columns
12. Perform filtering setting your own criteria based on your dataset
13. Perform slicing using index and column names
14. Perform slicing using index only
15. Check null values for each column separately
16. Display the total number null values (missing values) all columns
17. Identify columns having missed or null values and replace them with:
 - a. Mean of the column
 - b. Minimum value of the column
 - c. Maximum value of the column
18. Export the filled data frame to new CSV file in the format (filename_yourname.csv')
19. Using the file in question 18, do histogram, pie chart, bar chart, line graphs and other visualizations using Matplotlib.

Missing Values:

The dataset contains null values. Three fill strategies were applied and compared:

Fill with mean: Preserves the overall distribution.

Fill with minimum: Assigns the lowest valid observed value.

Fill with maximum: Assigns the highest valid observed value.

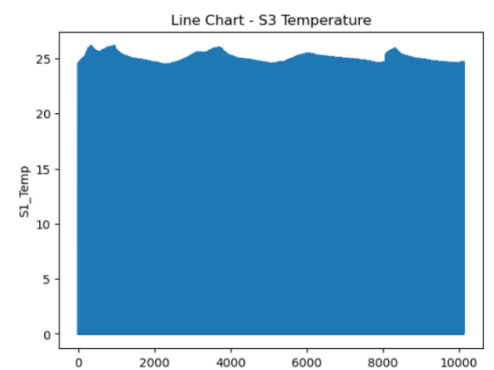
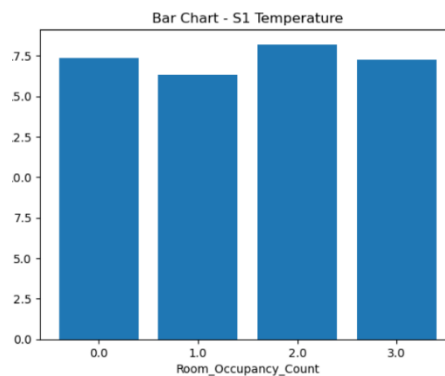
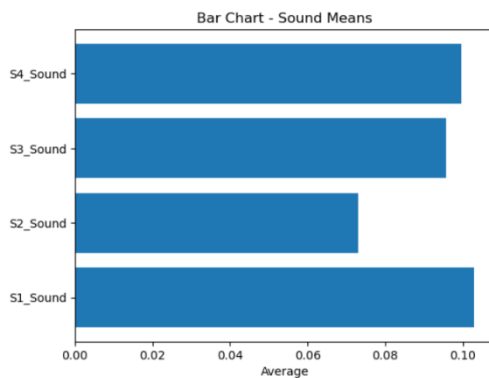
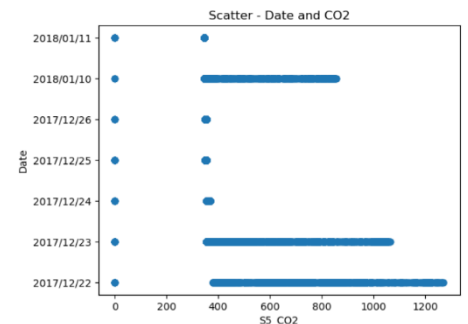
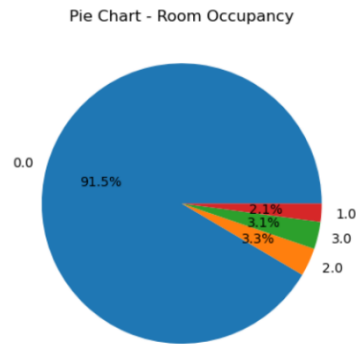
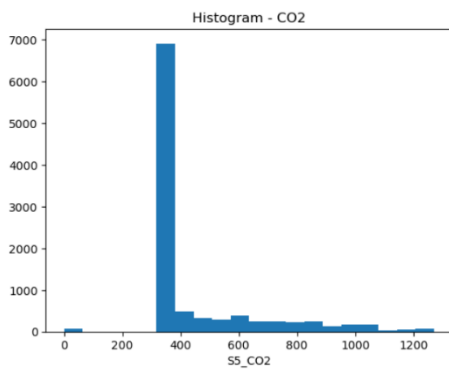
For the final output CSV, I use an additional step to replace all null values with zeros.

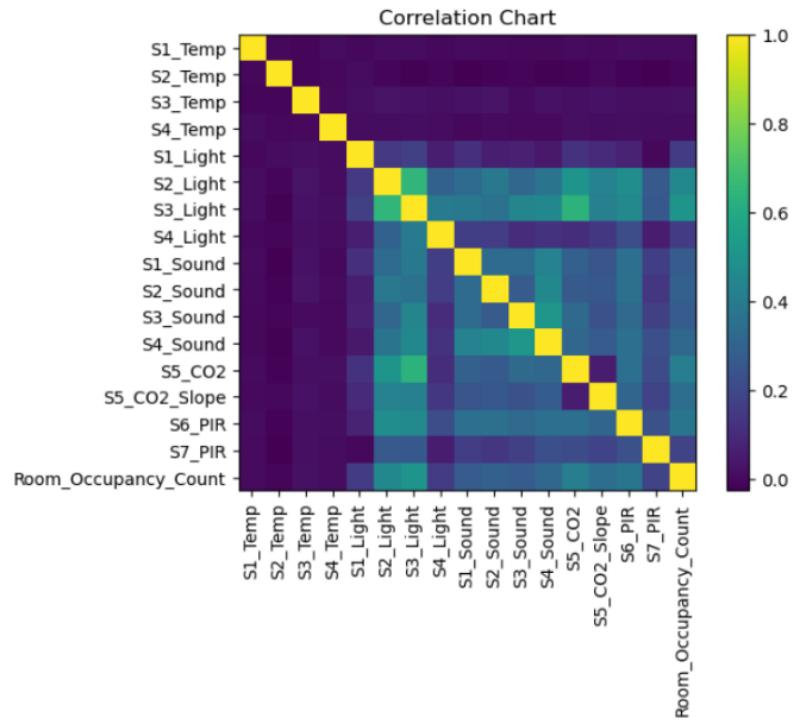
Grouping:

Group	Aggregation	Target Column
Room_Occupancy_Count	Mean	S1_Temp
Date	Median	S5_CO2_Slope
Date	Sum	Room_Occupancy_Count
Room_Occupancy_Count	Minimum	S5_CO2
Room_Occupancy_Count	Maximum	S5_CO2

Visualizations:

Chart Type	Data Used	Purpose
Histogram	S5_CO2	Observe the distribution of CO2 readings
Pie Chart	Room_Occupancy_Count	Show occupancy level proportions
Bar Chart	Mean S1_Temp by occupancy	Compare temperature across occupancy levels
Horizontal Bar Chart	Mean of sound sensors	Compare average sound levels across sensors
Line Graph	S3_Temp	Track temperature changes over time
Correlation Chart	Correlation matrix of numeric columns	Visualize correlations between all sensors
Scatter Plot	S5_CO2 and Date	Explore CO2 variation across dates





4. Output Files

Occupancy_Estimation_AlperenÇok.ipynb - Executed Jupyter Notebook containing all 19 questions with code and output.

Occupancy_Estimation_AlperenÇok.csv - Cleaned dataset with missing values filled by zero.

5. Conclusion

This project provided experience with the core features of the Pandas and Matplotlib libraries. The dataset presented a challenging missing value structure, which was addressed through three different fill strategies. Grouping and filtering operations revealed meaningful patterns between sensor readings and room occupancy levels. The visualizations further supported these observations by presenting the data in an accessible and interpretable format.